

Picosecond photoresponse in van der Waals heterostructures

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Two-dimensional crystals such as graphene and transition-metal dichalcogenides¹ demonstrate a range of unique and complementary optoelectronic properties^{2,3}. Assembling different two-dimensional materials in vertical heterostructures⁴ enables the combination of these properties in one device, thus creating multifunctional optoelectronic systems with superior performance. Here, we demonstrate that graphene/WSe₂/graphene heterostructures ally the high photodetection efficiency of transition-metal dichalcogenides^{5,6} with a picosecond photoresponse comparable to that of graphene^{7–9}, thereby optimizing both speed and efficiency in a single photodetector. We follow the extraction of photoexcited carriers in these devices using time-resolved photocurrent measurements and demonstrate a photoresponse time as short as 5.5 ps, which we tune by applying a bias and by varying the transition-metal dichalcogenide layer thickness. Our study provides direct insight into the physical processes governing the detection speed and quantum efficiency of these van der Waals heterostructures, such as out-of-plane carrier drift and recombination. The observation and understanding of ultrafast and efficient photodetection demonstrate the potential of hybrid transition-metal dichalcogenide-based heterostructures as a platform for future optoelectronic devices.

The optoelectronic response of two-dimensional crystals is currently the subject of intense investigation^{1–3,5–16}, prompted by the need for next-generation photodetectors with superior performance in terms of efficiency, detection speed, as well as flexibility and transparency¹⁷. High photon absorption^{5,18} and large photoconducting gain^{11,12,14} have been observed in devices based on semiconducting two-dimensional crystals. However, the observed response time typically ranges from nanoseconds¹⁶ to seconds^{11,14}, with faster devices often displaying lower responsivity¹¹. Therefore, the main challenge is to develop and assess photodetectors based on two-dimensional semiconductor crystals that simultaneously possess a large active area, high internal efficiency and fast response time.

A promising approach to create such versatile devices is to sandwich a transition-metal dichalcogenide (TMD) layer between two graphene sheets serving as charge extraction contacts. In contrast to lateral photodetectors, such vertical van der Waals heterostructures⁴ have the advantage of possessing a large, scalable active area and an atomically short charge extraction channel, potentially enabling both efficient and fast photodetection. Whereas the quantum efficiency of these van der Waals devices^{5,6,13} and the dynamics of photocarrier creation and relaxation in TMDs^{19–27} have been addressed, the response time of TMD-based photodetectors, as well as the dynamic processes governing their quantum efficiency, remain elusive.

Here, we report on the intrinsic processes that limit the performance of photodetectors based on high-quality G/WSe₂/G (where G is graphene) van der Waals heterostructures encapsulated in hexagonal boron nitride (hBN)²⁸. We perform time-resolved photocurrent measurements^{7,29} on devices consisting of WSe₂ flakes with a range of thicknesses (monolayer and multilayers from 2.2 to 40 nm). This technique, which combines electronic detection with subpicosecond optical excitation, allows probing of the extraction (Fig. 1a) and loss dynamics of the photoexcited charge carriers in the photoactive TMD layer. We use WSe₂ as the photoactive material because of its high optical quality^{15,30}, and monolayer graphene flakes, placed below and above the TMD layer, as source and drain electrodes with gate-tunable workfunctions (Fig. 1b). Figure 1c presents an optical image of a typical device comprising a WSe₂ flake with a thickness of 2.2 nm (three atomic layers). Unless otherwise specified, all measurements presented in Figs 1 and 2 are obtained from this particular device.

We first characterized the photoresponse of our devices at room temperature using a scanning photocurrent microscope (see Methods). The photocurrent map presented in Fig. 1d shows that the photosensitive region corresponds to the area where the three main layers (G/WSe₂/G) are superimposed. The photocurrent spectrum (Fig. 1e) measured in this area exhibits three main peaks (labelled A, B and A'), the positions of which match well with the exciton absorption peaks of WSe₂^{30–32}. This confirms that the photocurrent originates from the generation of photoexcited charge carriers following light absorption in the WSe₂ layer, followed by the transfer of these charges to the graphene electrodes. The polarity and efficiency of the photocurrent are driven by the potential drop across the WSe₂ layer (ΔV), which can be tuned by applying a bias (V_B) or gate (V_G) voltage^{5,6} (Supplementary Section I).

To investigate the dynamics of the photoexcited charges, we performed time-resolved photocurrent measurements^{7,29} by exciting the photodetector with a pair of ultrashort pulses (~ 200 fs, centred at 1.55 eV) separated by a variable time delay Δt (Fig. 1b, see Methods). Also known as photocurrent autocorrelation measurements—in analogy to autocorrelation measurements using nonlinear optical crystals—this technique exploits the nonlinear power dependence of the photocurrent to extract the photocurrent dynamics. Figure 2a shows that this nonlinear power dependence is significant for average laser pulse intensities exceeding 10 kW cm^{-2} . The observed sublinearity probably originates from saturable absorption^{22,33} (phase space filling), but we cannot exclude possible contributions from increased interactions between photocarriers (such as electron–hole recombination²⁹, exciton–exciton annihilation^{21,25} and carrier–carrier scattering²²) or from screening of the external bias (Supplementary Section II).

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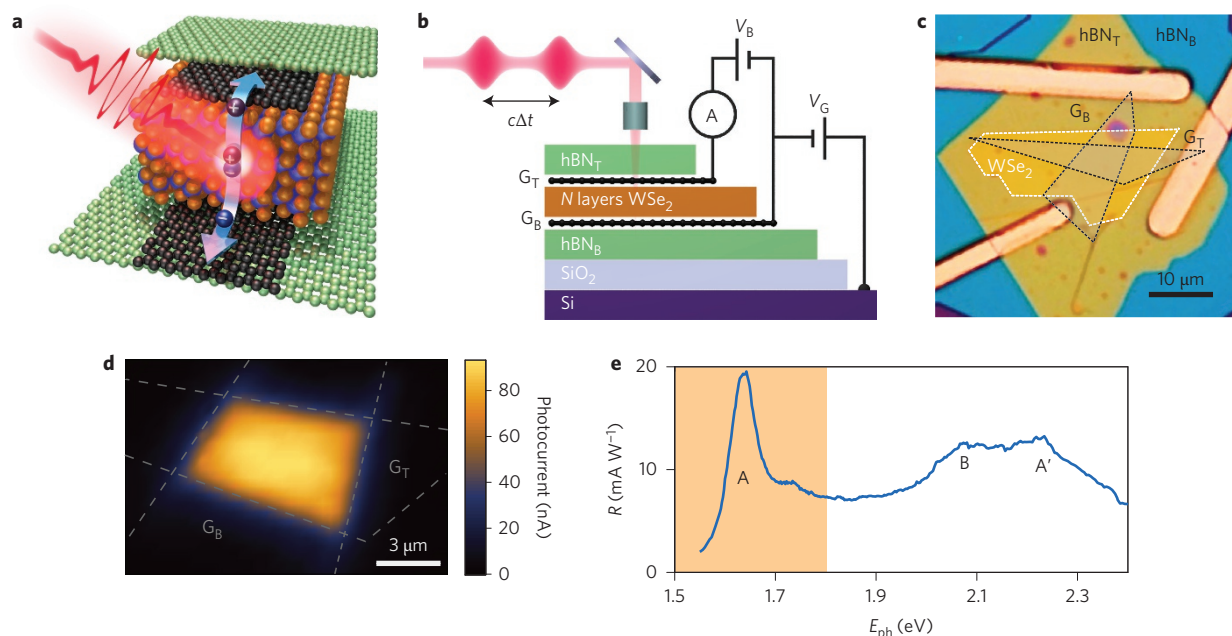


Figure 1 | Photocurrent generation in the G/WS₂/G heterostructure. **a**, Schematic representation of photoexcited charge carrier dynamics in a hBN/G/WS₂/G/hBN heterostructure. Graphene and hBN layers are coloured in black and green, respectively, and W and Se atoms in blue and orange. Following pulsed-laser excitation, excitons or electron-hole pairs are created, separated and transported to the graphene electrodes. **b**, Schematic illustrating the experimental time-resolved photocurrent set-up and cross-sectional view of the device. Two ultrashort pulses, delayed by a controllable distance $c\Delta t$ (c is the speed of light), are focused on the device, which contains a Si backgate to which a gate voltage (V_G) is applied between the top (G_T) and bottom (G_B) graphene layers through which the photocurrent is measured (see Methods). **c**, Optical image of a heterostructure comprising a 2.2-nm-thick WS₂ flake. The graphene and WS₂ flakes are outlined and shaded for clarity. **d**, Photocurrent map obtained by scanning a focused laser beam with a wavelength of $\lambda = 759$ nm and power $P = 5$ μW on the device in **c**, with $V_G = 0$ V and $V_B = 0.4$ V. The photocurrent is mainly generated in the region where graphene layers (delimited by grey dotted lines) overlap. **e**, Responsivity spectrum of the device shown in **c** measured at a constant power $P = 5$ μW, with $V_G = 0$ V and $V_B = 0.4$ V. Exciton peaks are labelled according to the convention of Wilson and Yoffe³¹ for WS₂ and confirm that the photocurrent stems from the vertical extraction of carriers generated in the WS₂ layer. Orange shading indicates the spectral range of the pulsed excitation (illustrated in **b**) used for time-resolved measurements.

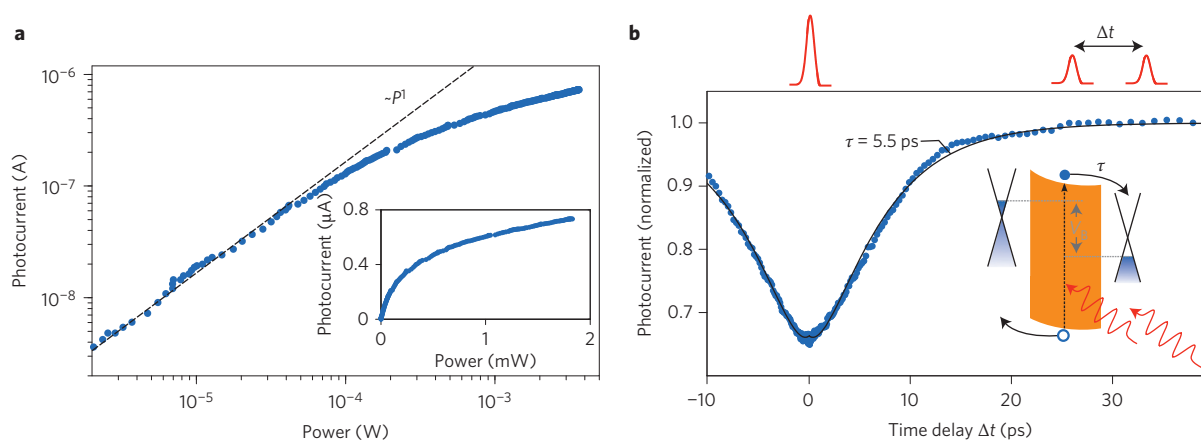


Figure 2 | Extraction of the photoresponse time of a G/2.2 nm WS₂/G heterostructure by time-resolved photocurrent measurements. **a**, Photocurrent versus laser pulse power P for $V_B = 0.2$ V. The transition from the linear to sublinear regime takes place at $P \approx 100$ μW (~ 10 kW cm⁻²). The dotted line corresponds to a linear relationship between photocurrent and P . Inset: the same data presented on a linear scale. **b**, Photocurrent as a function of time delay between two pulses (illustrated above the plot) with $P = 300$ μW, at $V_B = 1.2$ V (blue dots). The photocurrent is normalized by the value of the photocurrent saturating at long Δt . The solid black line is a fit to the data using the model described in the Methods, yielding a time constant $\tau = 5.5 \pm 0.1$ ps. Inset: schematics of the photoresponse time in a G/WS₂/G heterostructure, here represented by a band diagram with a bias voltage V_B applied between the graphene layers. Red sinusoidal arrows, the blue filled circle and blue open circle symbolize photons, and photoexcited electron and hole, respectively. The photoresponse time τ , which corresponds to the time that photocarriers reside in the photoactive area, is represented by black arrows.

As a result of this sublinearity, photocurrent autocorrelation measurements, such as the one presented in Fig. 2b, exhibit a symmetric dip around $\Delta t = 0$. We ascribe the dynamics of the dip to

depletion of photocarriers from the active region²⁹, which we can describe (see Methods) with a characteristic time constant τ that corresponds to the photoresponse time of our WS₂-based

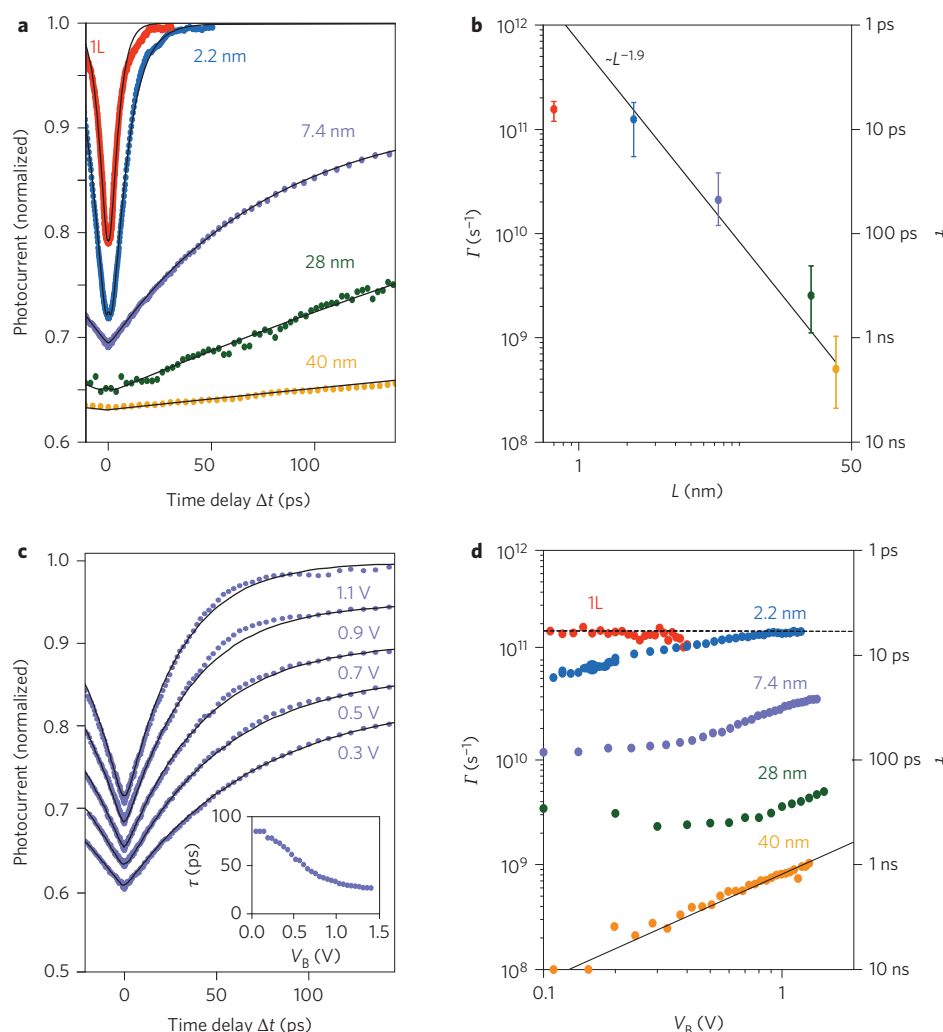


Figure 3 | Tuning of photoresponse time τ by variation of the WSe₂ layer thickness L and bias voltage V_B . **a**, Time-resolved photocurrent measurements on heterostructures with different L (curves are offset for clarity). All measurements are obtained using a laser pulse power of $P \approx 300 \mu\text{W}$ and bias voltage $V_B = 0.5 \text{ V}$, except for the monolayer (1L), which was measured at $V_B = 0 \text{ V}$. Solid black lines are fits to the data. **b**, Photoresponse rate $\Gamma = 1/\tau$ versus L . Data points represent the averaged values of Γ for different V_B (typically from 0.1 to 1 V) and error bars correspond to the minimum and maximum values of Γ . The solid black line is a power-law fit (excluding the monolayer device), yielding $\Gamma \propto L^{-1.9 \pm 0.3}$. **c**, Time-resolved photocurrent measurements on a heterostructure with $L = 7.4 \text{ nm}$ at various bias voltages V_B and $P = 300 \mu\text{W}$ (curves are offset for clarity). Inset: photoresponse time τ extracted from this heterostructure as a function of V_B . **d**, Photoresponse rate Γ as a function of V_B for various values of L . The solid black line corresponds to a linear relationship between Γ and V_B , whereas the dotted line shows the effective minimum response time $\tau = 5.5 \text{ ps}$.

photodetector. We will show that, regardless of the mechanism that leads to the sublinear power dependence, the extracted time constant is representative of the photophysics that takes place in the device.

The value of τ was determined for five devices with different WSe₂ layer thicknesses L (Fig. 3a), and was found to increase with L and vary by more than two orders of magnitude: from 10 ns (40 nm device) to 5.5 ps (monolayer and 2.2 nm devices). Figure 3b also shows that τ scales quadratically with thickness L (except the monolayer, which we discuss later). The highest response rate of $\Gamma = \tau^{-1} = (5.5 \pm 0.1 \text{ ps})^{-1}$, corresponding to a bandwidth of $f = 0.55/\tau = 100 \text{ GHz}$, is comparable to the intrinsic photo-switching speed of typical graphene detectors ($\sim 260 \text{ GHz}$)⁷. We note that, in general, ultrafast devices may have an operating speed limited by their resistance-capacitance (RC) time. In the present devices the relevant capacitance is that of the WSe₂ channel, and the resistance is the sum of the graphene resistance and graphene-metal contact resistance. We indeed expect the photoresponse time to be RC-limited to $\tau > 0.4 \text{ ns}$ for the thinnest

devices ($L < 10 \text{ nm}$, Supplementary Section III). Using real-time electronic measurements, we find an instrument-limited higher bound of $\sim 1.6 \text{ ns}$. This is already significantly faster than any TMD-based photodetectors reported to date^{6,11,12,14} and can be improved further by optimizing the circuit design, graphene resistance and contact resistance, up to the intrinsic limits that are reported in this work.

To address the underlying physics governing the dynamics and efficiencies in more detail, we studied the dependence of τ on bias voltage V_B (Fig. 3c) and observe that τ decreases with V_B (Fig. 3c, inset), indicating that photocarriers escape the active region more quickly as the electric field across the WSe₂ layer increases. For devices made of a relatively thick WSe₂ layer ($L \geq 7.4 \text{ nm}$), the response rate Γ depends linearly on V_B (Fig. 3d). This observation and the scaling of τ with L^2 suggests out-of-plane drift-diffusive transport, where, to reach the graphene contact, photocarriers generated in the middle of the WSe₂ layer travel a distance $L/2$ with a drift velocity $v_d = \mu E$ where μ is the out-of-plane photocarrier mobility and E is the electric field across the WSe₂ layer. Assuming that

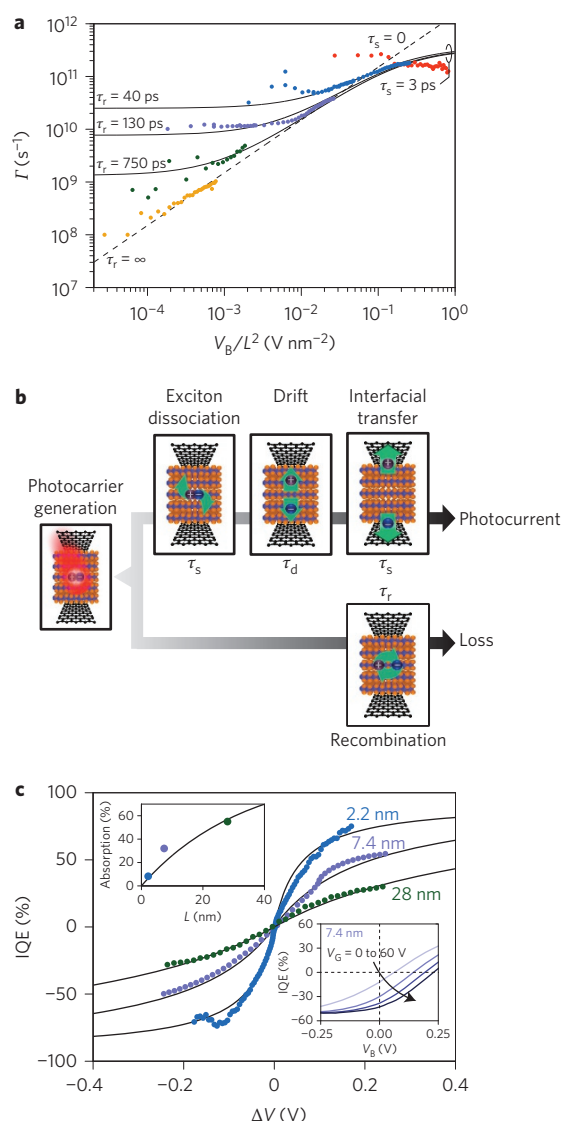


Figure 4 | Dynamic processes governing the photoresponse of G/WSe₂/G heterostructures. **a**, Response rate $\Gamma = 1/\tau$ versus V_B/L^2 (same data as Fig. 3d). The dotted line corresponds to the diffusive transport model discussed in the main text with photocarrier mobility $\mu = 0.010 \pm 0.003$ cm² V⁻¹ s⁻¹. The uncertainty in μ is estimated from the small variations in the mobility of each device, which, among other things, could stem from extrinsic factors (for example, unintentional doping) or variations in the laser power used to measure each device (Supplementary Section VII). Solid black lines correspond to the equation $\tau^{-1} = (\tau_d + \tau_s)^{-1} + \tau_r^{-1}$ with $\mu = 0.01$ cm² V⁻¹ s⁻¹, $\tau_s = 3$ ps and the values of τ_r indicated for each line. **b**, Schematic of the processes contributing to the response time. Radiative and non-radiative photocarrier recombination (τ_r) leads to losses, whereas photocurrent is generated by photocarrier drift (τ_d) in combination with exciton dissociation and/or charge transfer at the G/WSe₂ interface (τ_s). **c**, IQE as a function of potential drop across the WSe₂ layer ΔV for devices with $L = 2.2$, 7.4 and 28 nm. Black lines correspond to the expression $\tau_r/(\tau_d + \tau_s + \tau_r)$, where $\tau_d = L^2/2\mu\Delta V$, using the values indicated in **a**. The experimental IQE values (data points) are obtained by measuring the photocurrent generated at low power ($P = 1$ μ W) with a quasi-continuous laser ($\lambda = 759$ nm) and by using the absorption as a fit parameter. Top left inset: fitted absorption values, which compare well with the absorption expected from Beer-Lambert's law (solid black line) using the bulk absorption coefficient of WSe₂ (3×10^5 cm⁻¹)³². Bottom right inset: IQE versus V_B measured on the 7.4-nm-thick device under the same illumination conditions as the main figure, with V_G ranging from 0 to 60 V in steps of 20 V.

$E \approx V_B/L$ (Supplementary Section IV), the time τ_d it takes for photocarriers to drift out of the active region is $\tau_d = L^2/2\mu V$. This simple expression, represented by the dotted line in Fig. 4a, captures the trend of Γ at low V_B/L^2 ($< 10^{-2}$ V nm⁻²), yielding $\mu = 0.010 \pm 0.003$ cm² V⁻¹ s⁻¹. This value is similar to the one reported recently for thick MoS₂ flakes obtained from transport measurements³⁴ and is consistent with the strong conductivity anisotropy observed in TMDs (the in-plane conductivity is typically two to three orders of magnitude larger than the out-of-plane conductivity)³⁵.

Interestingly, for higher values of V_B/L^2 (> 0.1 V nm⁻²) the response rate reaches a limit. For $L = 2.2$ nm (trilayer), this occurs for high V_B ($V_B > 0.6$ V), but the monolayer device reaches this limit without any applied bias. This saturation behaviour indicates the existence of an additional process that occurs in series with the drift process, thus prolonging the carrier extraction process. We find that the timescale τ_s of this additional process is bias-independent and is ~ 3 – 5 ps. This value is similar to the transfer time of photocarriers at the interface between WSe₂ and graphene (~ 1 ps)²⁰ and to the hot exciton dissociation time in few-layer MoS₂ (~ 0.7 ps)²⁷, which were both recently measured in all-optical pump-probe experiments. We thus argue that the response rate at high V_B/L^2 has a higher bound that is determined by these two processes rather than by drift (Fig. 4b), and this constitutes an intrinsic limitation for this device geometry.

We also observe a lower bound for the response rate, occurring at low V_B , which we attribute to a loss mechanism that occurs in parallel to the photocurrent generation process (Fig. 4b). We find that this process presents a strong thickness dependence, with time constant τ_r ranging from 40 ps (2.2 nm device) up to > 10 ns (40 nm device). We attribute this effect to radiative and non-radiative recombination of photocarriers^{19,23,24,26} and energy transfer via dipole-dipole interaction to the graphene sheets³⁶, two processes that have been shown to depend on TMD layer thickness. The latter interpretation is corroborated by the quenching of the WSe₂ photoluminescence observed in the trilayer (tenfold quenching) and monolayer (300-fold quenching) devices (Supplementary Section V).

We now show that the internal quantum efficiency (IQE, defined as the ratio between the number of extracted photocarriers and the number of absorbed photons) of the photodetector is the direct result of the competition between the photocarrier extraction time ($\tau_d + \tau_s$) and recombination loss (τ_r). Figure 4c shows the IQE as a function of ΔV (the potential drop across the WSe₂ layer), which can be controlled by both V_G and V_B (Fig. 4c, bottom right inset). We find that the experimental IQE matches well the extraction efficiency $\tau_r/(\tau_d + \tau_s + \tau_r)$ derived from the dynamic model shown in Fig. 4b. This concordance confirms that the dynamic processes identified in our time-resolved study correspond to the relevant physical mechanisms that govern the photoresponse of the device in the linear response regime (low-power, quasi-continuous excitation). It further demonstrates that the high IQEs observed in TMD-based vertical photodetectors ($> 85\%$)⁶ arise from the short extraction time of the photocarriers out of the thin TMD channel, thereby outcompeting the loss mechanisms.

This suggests that the IQE can be optimized by reducing the channel length, and thus minimizing the TMD thickness. Interestingly, for the most extreme case of a WSe₂ monolayer, we observe a very low IQE ($\sim 6\%$, assuming 5% absorption) despite a very short response time. This indicates that the observed dynamics in this device, as opposed to thicker ones, correspond to the losses, rather than to the extraction of charges. This is supported by the observation that the response time is bias-independent. The two loss mechanisms mentioned previously (intrinsic recombination and energy transfer) should indeed both occur much faster for a monolayer: recombination is facilitated by the direct nature of the bandgap of monolayer WSe₂^{19,30} and loss through energy transfer exhibits a strong power-law dependence with the inverse dipole-dipole distance³⁶.

We conclude that G/WSe₂/G devices made of trilayer WSe₂ (2.2 nm) offer the best compromise for optimizing the IQE as they exhibit both a fast photoresponse (down to 5.5 ps) and a high IQE (>70%). Moreover, the device can be efficiently operated without dark current and thus with a low noise level, by applying a large gate voltage and no bias voltage (noise-equivalent power down to $\sim 5 \times 10^{-12}$ W Hz^{-1/2}, Supplementary Section VI). Further improvement of the external quantum efficiency (EQE = 7.3% in our 2.2-nm-thick device) can be achieved by enhancing the light-matter interaction in the TMD layer with, for instance, optical waveguides^{9,11} and cavities or plasmonic nanostructures^{5,10}. Stacking multiple semiconducting two-dimensional crystals with different bandgaps is also a promising way to extend the spectral absorption range² without compromising the photoresponse time much. Our comprehensive understanding of the photocurrent dynamics in G/TMD/G heterostructures paves the way for designing van der Waals heterostructures for fast and efficient optoelectronic applications, such as high-speed integrated data communication systems.

Methods

Methods and any associated references are available in the [online version of the paper](#).

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Author contributions

M.M. and F.H.L.K. conceived and designed the experiments. M.M., P.S. and F.V. fabricated the samples, carried out the experiments and performed the data analysis. K.W. and T.T. provided boron nitride crystals. K.G.S. and A.R.P. provided assistance for the photoluminescence measurements. M.M., F.V., K.J.T., P.S. and F.H.L.K. co-wrote the manuscript, with the participation of K.G.S. and A.R.P.

Additional information

Supplementary information is available in the [online version](#) of the paper. Reprints and permissions information is available online at www.nature.com/reprints. Correspondence and requests for materials should be addressed to F.H.L.K.

Competing financial interests

The authors declare no competing financial interests.

Methods

Device fabrication. The van der Waals heterostructures were prepared using a layer assembly technique similar to that reported by Wang and co-workers²⁸. hBN was first exfoliated onto a polymer (poly(methyl methacrylate) or polypropylene carbonate), and one of the flakes was used to successively pick up graphene, WSe₂, graphene and hBN flakes from the substrate (oxidized silicon). The hBN flakes were typically 15–50 nm thick, depending on the device. The resulting stack was deposited onto a degenerately doped silicon substrate covered with a 285-nm-thick SiO₂ layer, which was used as a back gate (V_G). Top and bottom graphene flakes were electrically connected by one-dimensional contacts³³ made of 2 nm Ti/100 nm Au.

Optoelectronic measurements. Photocurrent was generated by focusing a laser beam (close to the diffraction limit) with a microscope objective (Olympus LUCPlanFLN $\times 40$) onto the device, and was measured with a preamplifier and a lock-in amplifier synchronized with a mechanical chopper. A supercontinuum laser (NKT Photonics SuperK Extreme) with tunable wavelength (from 500 to 1,500 nm), pulse duration of ~ 40 ps and repetition rate of 40 MHz was used to characterize the devices. Time-resolved photocurrent measurements were performed with ~ 200 fs pulses (at the sample, not Fourier-transform-limited) centred at 800 nm (1.55 eV) with a spectral bandwidth of ~ 200 nm. Because the laser spectrum overlaps with the A exciton peak (Fig. 1e), we expect a significant contribution from these excitons to

the photocurrent generated with this laser. Laser pulses were generated by a Ti:sapphire laser (Thorlabs Octavius) with a repetition rate of 85 MHz. The optical beam was split into two arms and recombined using 50/50 beamsplitters. One arm contained a motorized translation stage that allowed for the generation of a computer-controlled time delay Δt between the two pulses.

Extraction of the photoresponse time. Global fitting of the time-resolved photocurrent (PC) signals was performed using the equation

$$\frac{PC(\Delta t)}{PC(\Delta t \rightarrow \infty)} = 1 - A \exp\left(\frac{-|\Delta t|}{\tau}\right) + B \exp\left(\frac{-|\Delta t|}{\tau_G}\right)$$

where amplitudes A and B , and time constants τ and τ_G , are the fitting parameters. The exponential time constant τ_G is an empirical term introduced to reproduce the flattening of the signals at low time delay ($|\Delta t| \leq 2.2$ ps), which is observed on the thinnest devices ($L < 5$ nm). This contribution is always small compared with the other exponential term ($|A| \gg |B|$), and slightly faster ($\tau_G \approx 2$ –5 ps). We attribute this signal to the optoelectronic response of the top and bottom graphene layers⁷. We emphasize that this small correction to the global exponential decay is introduced for the sake of generality of the fitting procedure and in any case does not influence the extracted response time τ and subsequent conclusions.